Credit Card Customer Churn Prediction

End-to-end analysis from exploratory data analysis and customer segmentation through model comparison, threshold optimization, and retention targeting.

10,127 TOTAL CUSTOMERS	**16.1%** CHURN RATE	**LightGBM** BEST MODEL
0.993 ROC-AUC	**0.91** F1 SCORE	**97%** ACCURACY

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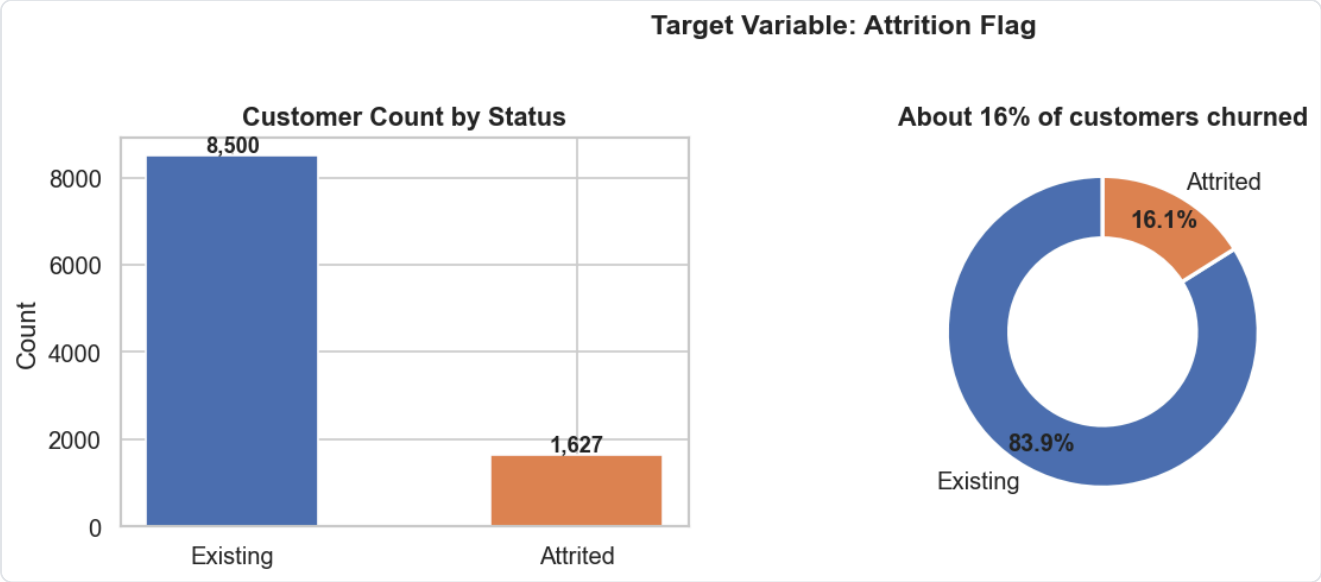
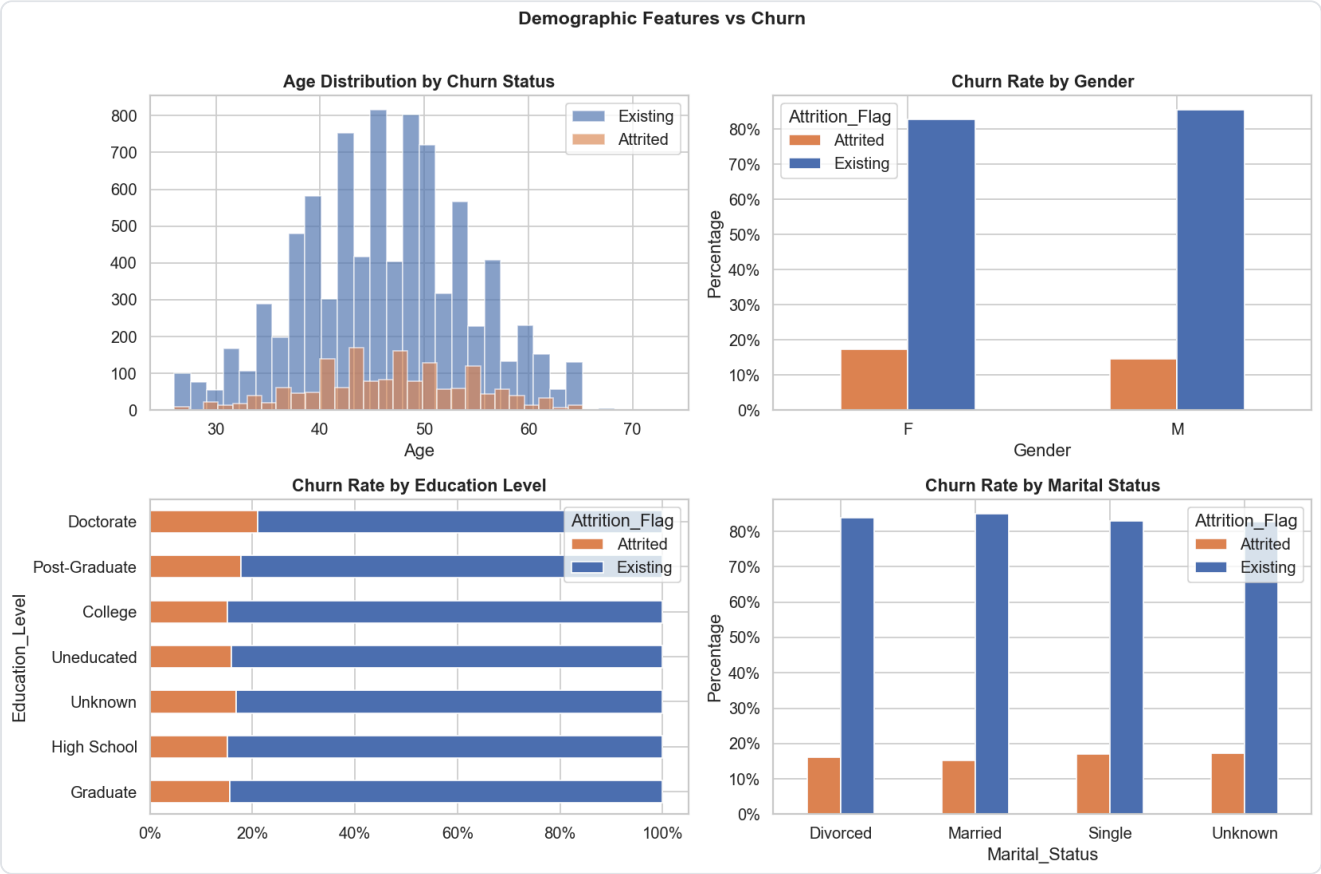
1. Exploratory Data Analysis

The dataset has 10,127 credit card customers with 21 features. About 16% closed their accounts. The following charts explore what separates churners from loyal customers.

Target Distribution

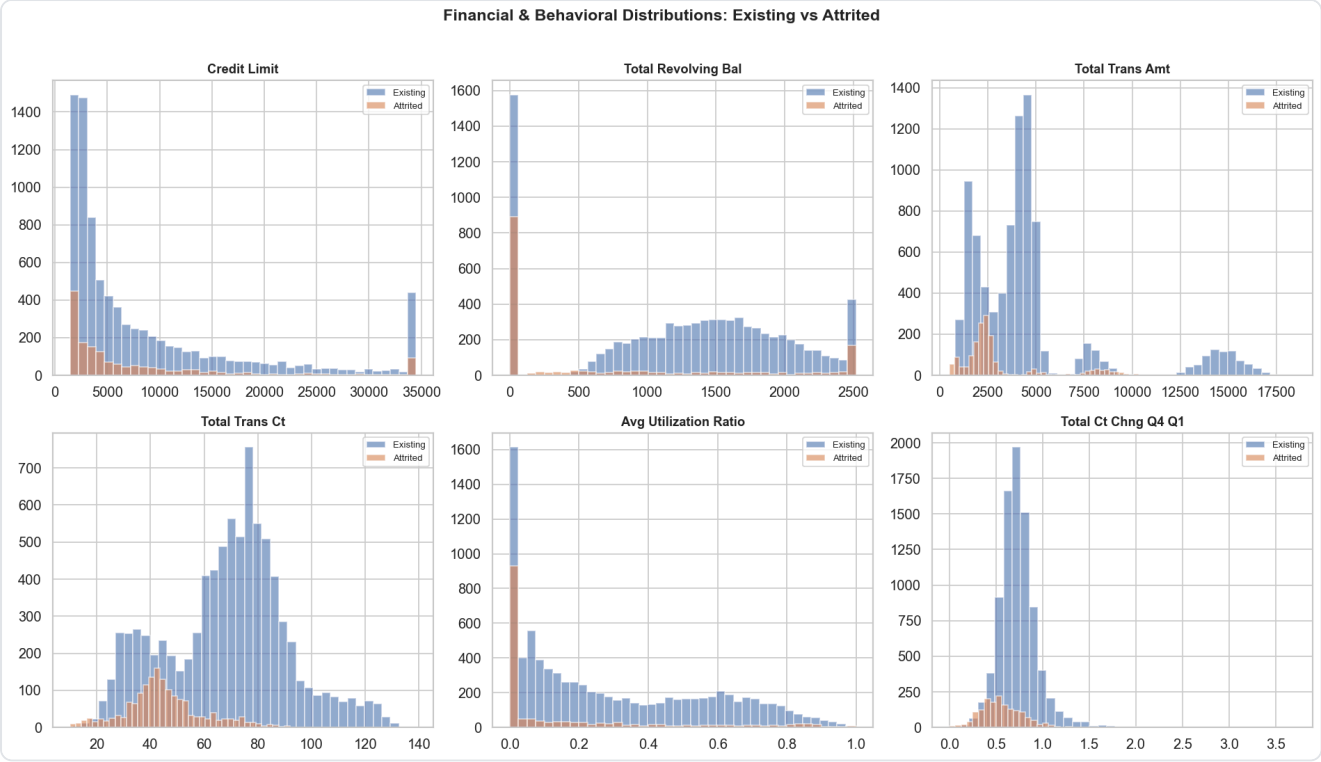


Demographics



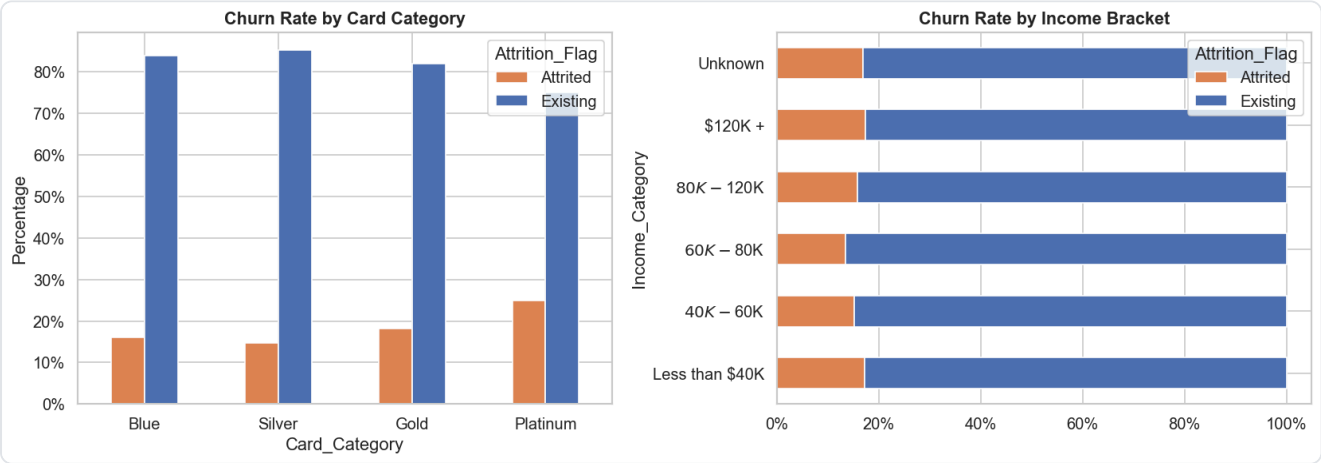
Churn rates are flat across gender, education, and marital status. None of these demographics differentiate churners in any meaningful way.

Financial and Behavioral Features

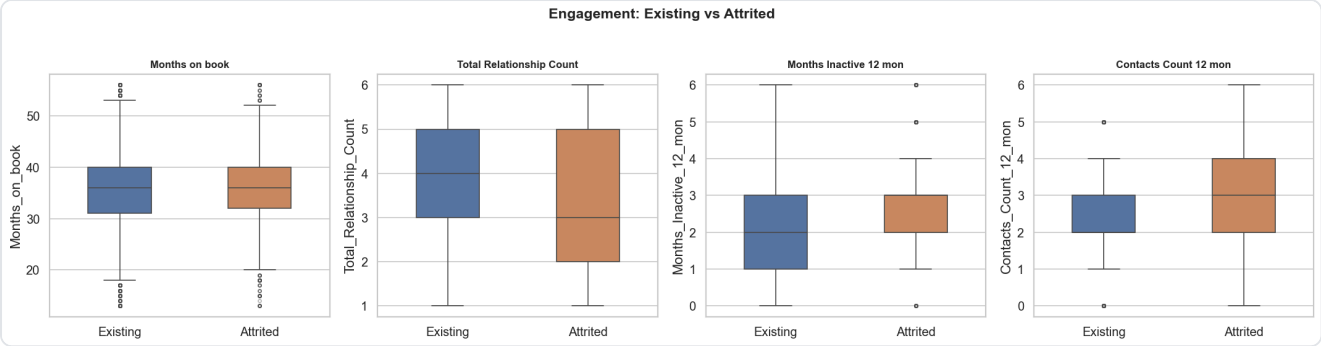


This is where separation shows up. Churners pile up on the low end of transaction count, amount, revolving balance, and utilization. They were already disengaging before they left.

Card Category and Income

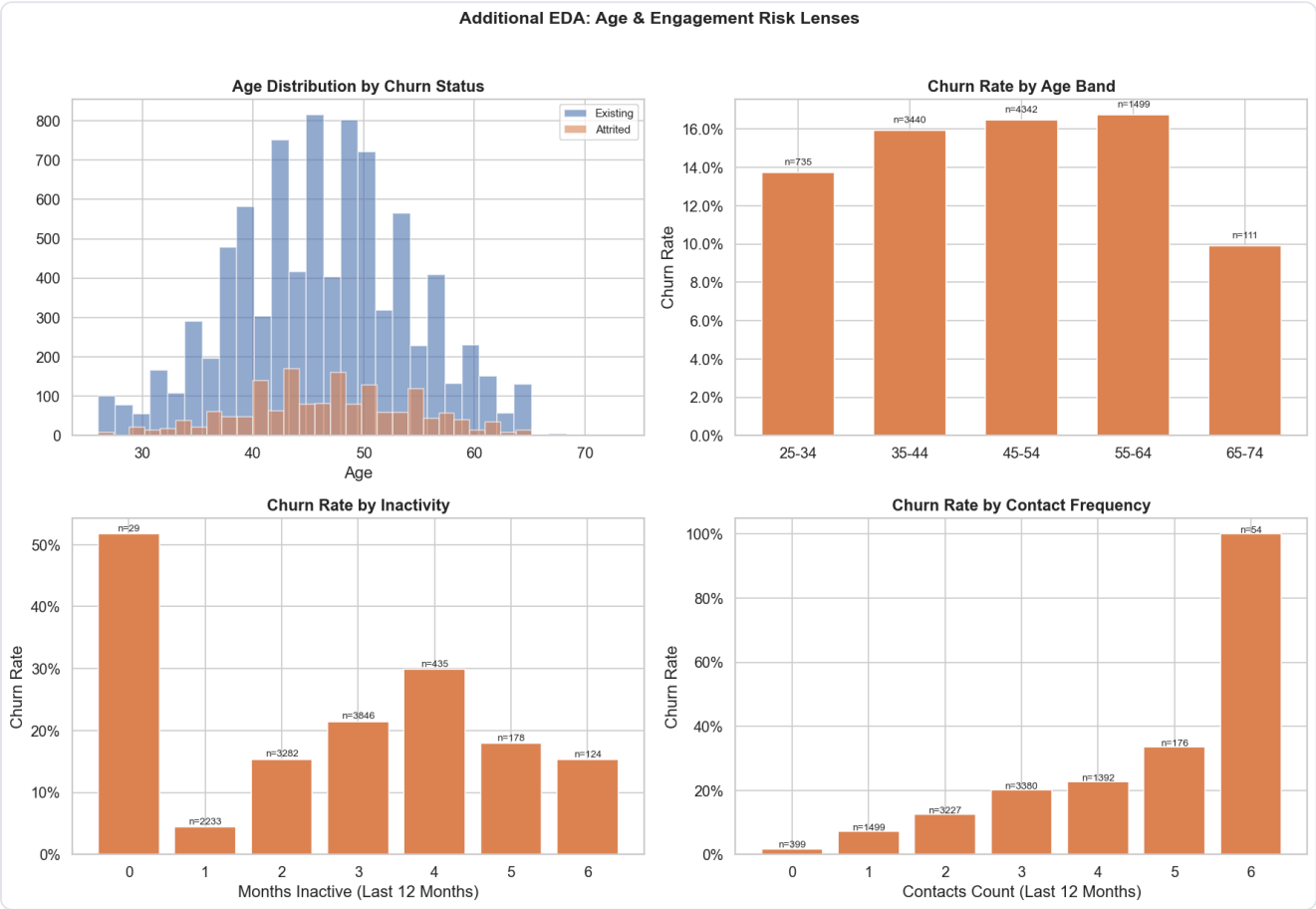


Customer Engagement



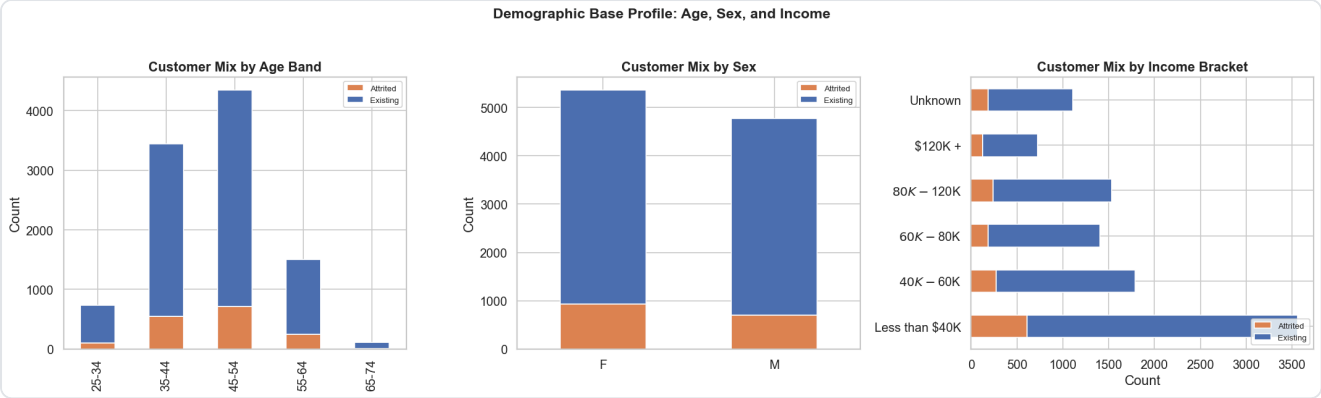
More months inactive, more bank contacts, fewer products held. Classic disengagement.

Additional Risk Views

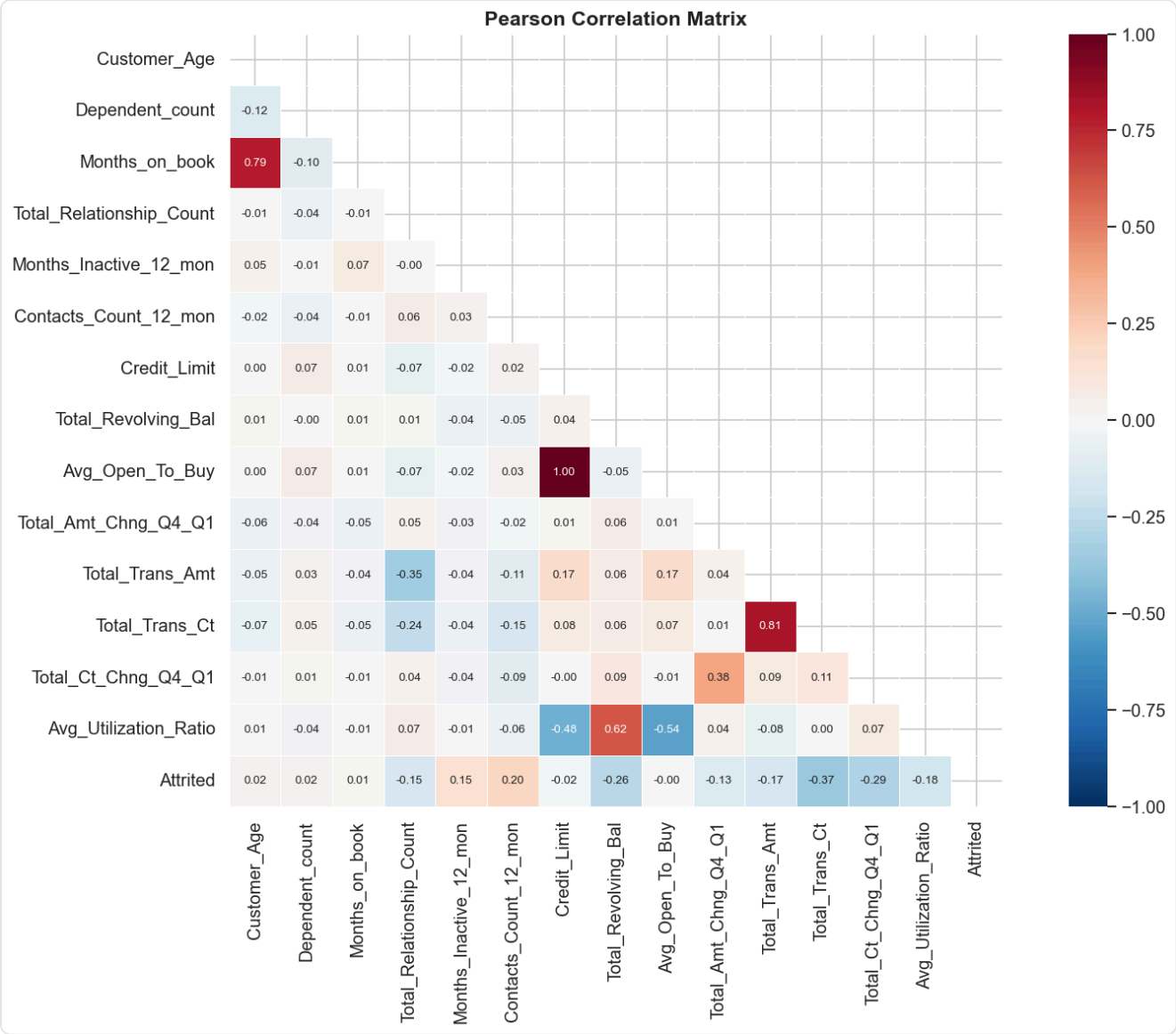


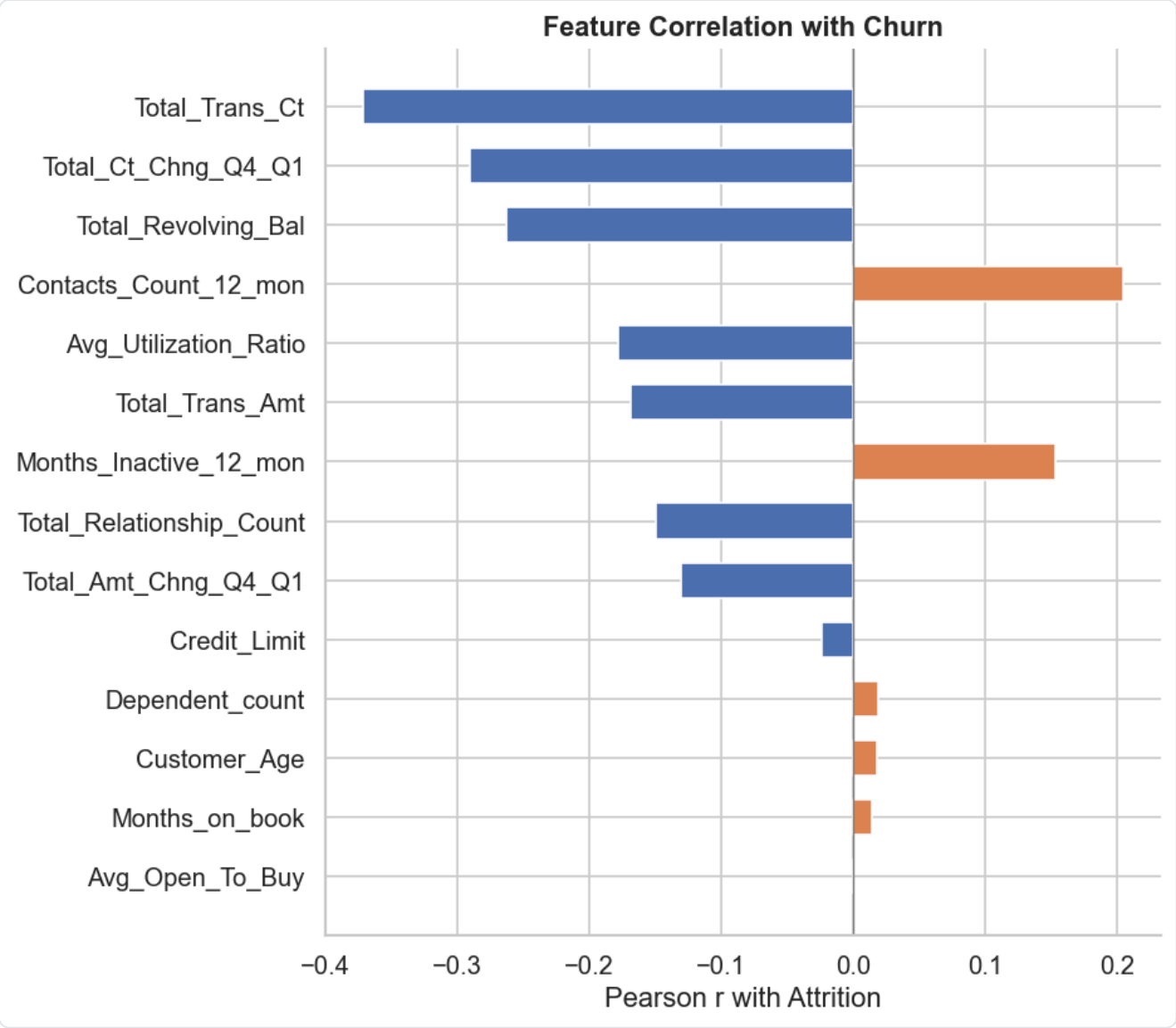
Churn jumps sharply with inactivity and repeated bank contacts, making both solid early-warning signals.

Demographic Base Profile



Correlation Analysis

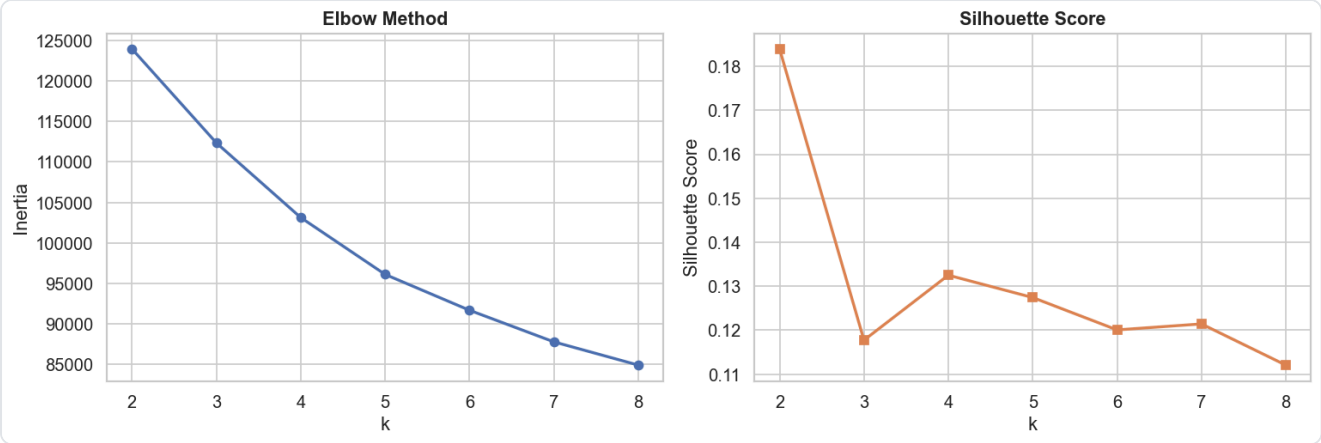

Pearson Correlation Matrix
Customer_Age
Dependent_count
Months_on_book
Total_Relationship_Count
Months_Inactive_12_mon
Contacts_Count_12_mon
Credit_Limit
Total_Revolving_Bal
Avg_Open_To_Buy
Total_Amt_Chng_Q4_Q1
Total_Trans_Amt
Total_Trans_Ct
Total_Ct_Chng_Q4_Q1
Avg_Utilization_Ratio
Attrited
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00



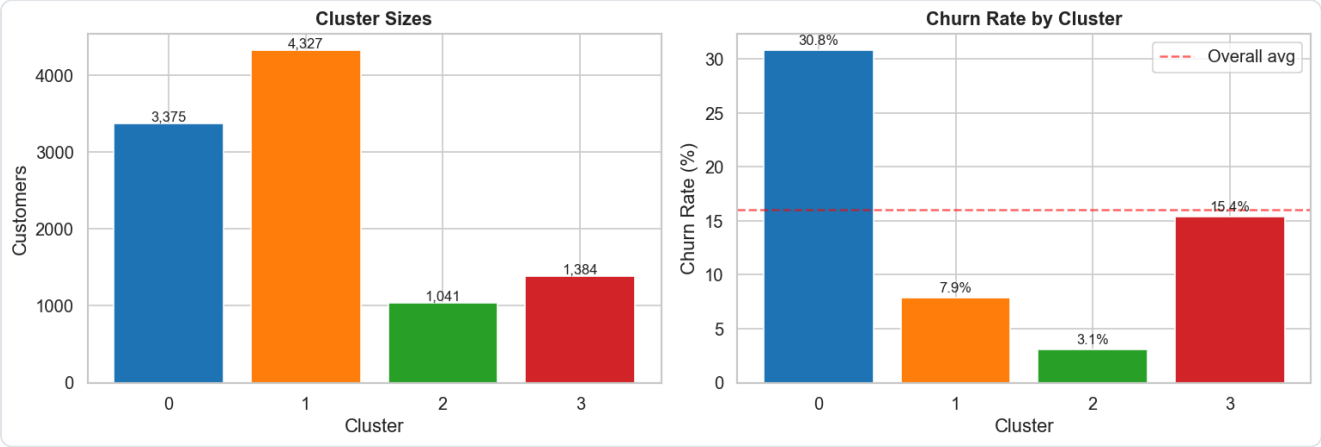
Transaction count and amount are the top two churn predictors by a wide margin, followed by the Q4/Q1 change ratio and revolving balance.

2. Customer Segmentation

K-Means clustering on scaled numeric features, with k=4 chosen for actionable granularity (elbow + silhouette analysis below).



Churn Rate by Cluster



One cluster has a 30.8% churn rate, nearly double the overall average. That is the at-risk segment.

Cluster Profiles

Cluster Profiles (feature means)

CLUSTER	0	1	2	3
CUSTOMER_AGE	46.700000	46.300000	45.300000	46.400000
CREDIT_LIMIT	5981.100000	3910.000000	13122.900000	26481.100000
TOTAL_TRANS_CT	57.700000	62.500000	104.700000	59.700000
TOTAL_TRANS_AMT	3267.400000	3510.500000	12821.100000	3638.600000
TOTAL_REVOLVING_BAL	403.500000	1683.900000	1393.700000	1211.700000
AVG_UTILIZATION_RATIO	0.100000	0.500000	0.200000	0.100000
MONTHS_INACTIVE_12_MON	2.500000	2.300000	2.200000	2.300000

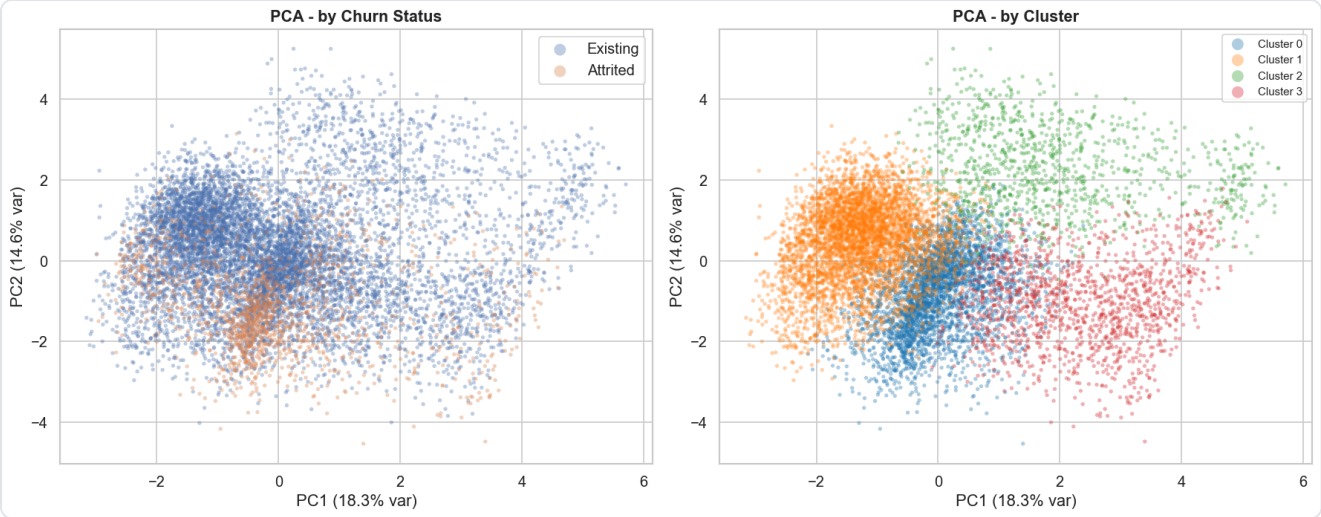
CLUSTER	0	1	2	3
TOTAL_RELATIONSHIP_COUNT	3.900000	4.100000	2.200000	4.000000
TOTAL_CT_CHNG_Q4_Q1	0.600000	0.800000	0.700000	0.700000
CHURN RATE (%)	30.800000	7.900000	3.100000	15.400000
SIZE	3375.000000	4327.000000	1041.000000	1384.000000

Feature Distributions by Cluster

3. Dimensionality Reduction

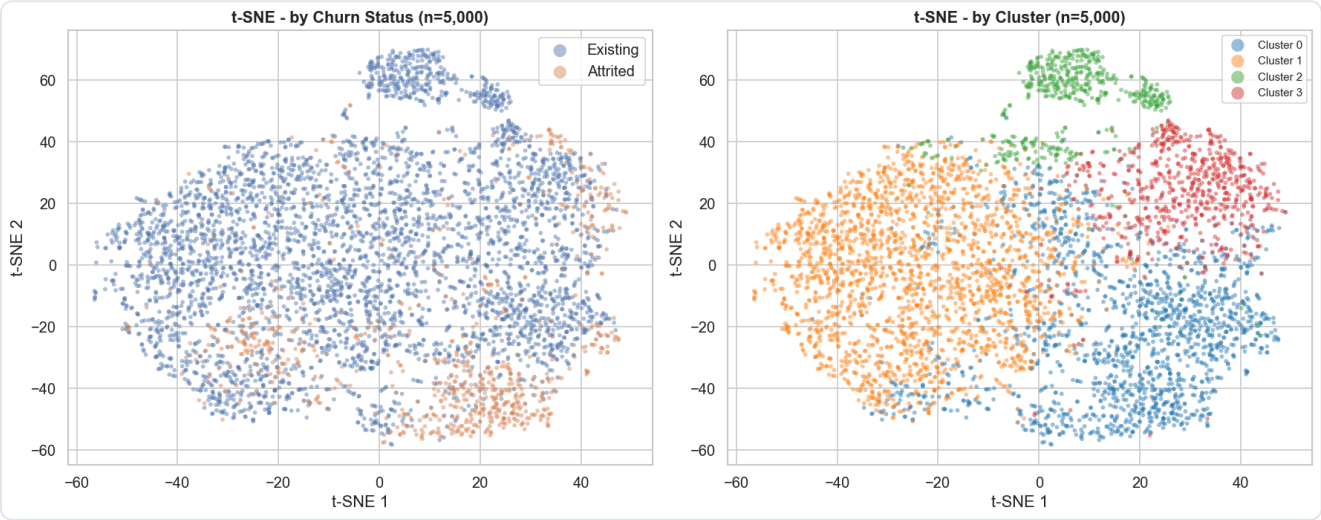
PCA and t-SNE project 14 features into 2D for visual inspection.

PCA Projection



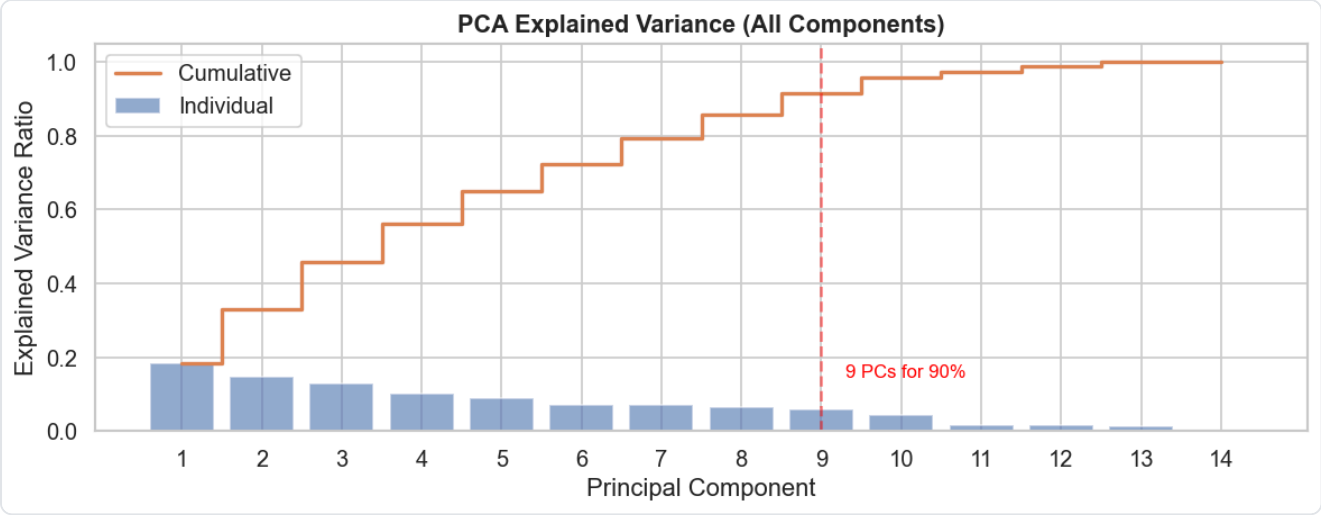
Churners concentrate in the lower-left, which maps to the high-churn cluster from K-Means.

t-SNE Projection



t-SNE pulls apart local structure more aggressively. Churners form their own pockets rather than being evenly distributed.

Explained Variance



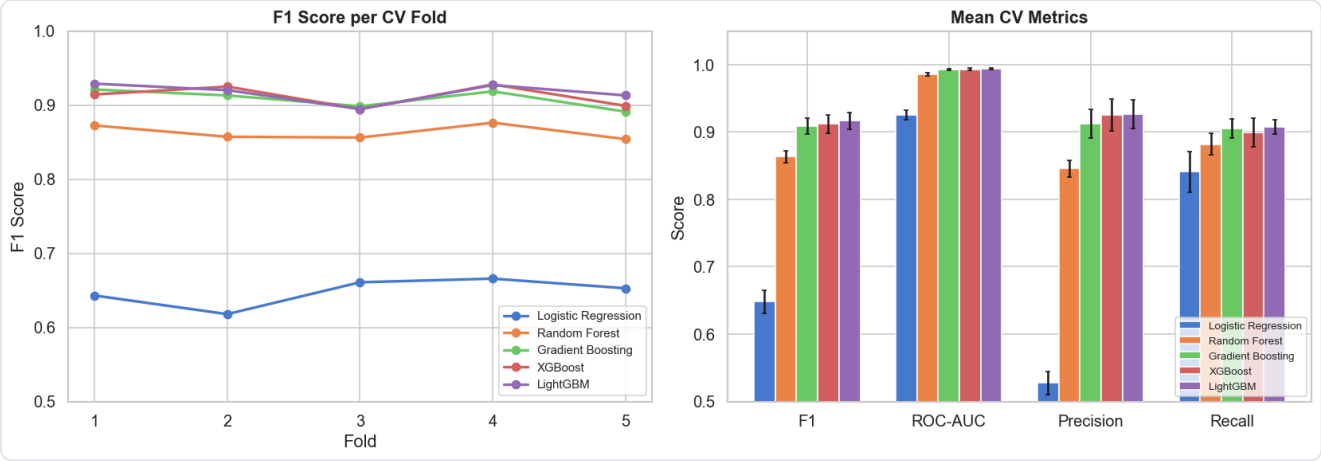
4. Model Training and Comparison

Five classifiers compared with stratified 5-fold cross-validation, all wrapped in pipelines with SMOTE.

```
Logistic Regression... F1=0.648 +/- 0.017  AUC=0.925
  Random Forest... F1=0.863 +/- 0.009  AUC=0.986
  Gradient Boosting... F1=0.908 +/- 0.012  AUC=0.992
  XGBoost... F1=0.912 +/- 0.013  AUC=0.993
  LightGBM... F1=0.917 +/- 0.012  AUC=0.993

Done
```

5-Fold CV Results (Training Set)

MODEL	F1 (MEAN +/- STD)	ROC-AUC (MEAN +/- STD)	PRECISION	RECALL
LOGISTIC REGRESSION	0.648 +/- 0.017	0.925 +/- 0.007	0.527	0.841
RANDOM FOREST	0.863 +/- 0.009	0.986 +/- 0.003	0.846	0.882
GRADIENT BOOSTING	0.908 +/- 0.012	0.992 +/- 0.001	0.912	0.905
XGBOOST	0.912 +/- 0.013	0.993 +/- 0.002	0.926	0.899
LIGHTGBM	0.917 +/- 0.012	0.993 +/- 0.001	0.927	0.907



5. Evaluation on Held-Out Test Set

Test Metrics

Test Set Performance

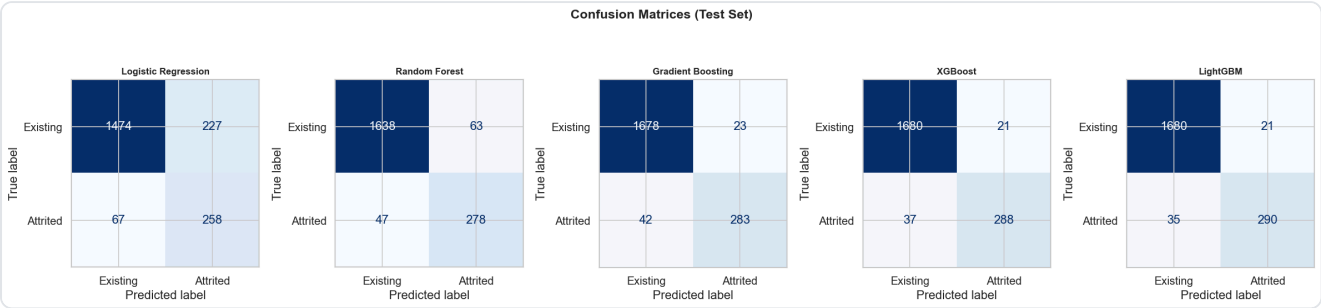
MODEL	F1	ROC-AUC	PRECISION	RECALL
LIGHTGBM	0.911900	0.992900	0.932500	0.892300
XGBOOST	0.908500	0.992400	0.932000	0.886200
GRADIENT BOOSTING	0.897000	0.991200	0.924800	0.870800
RANDOM FOREST	0.834800	0.982000	0.815200	0.855400
LOGISTIC REGRESSION	0.637000	0.915700	0.532000	0.793800

Best Model: LightGBM

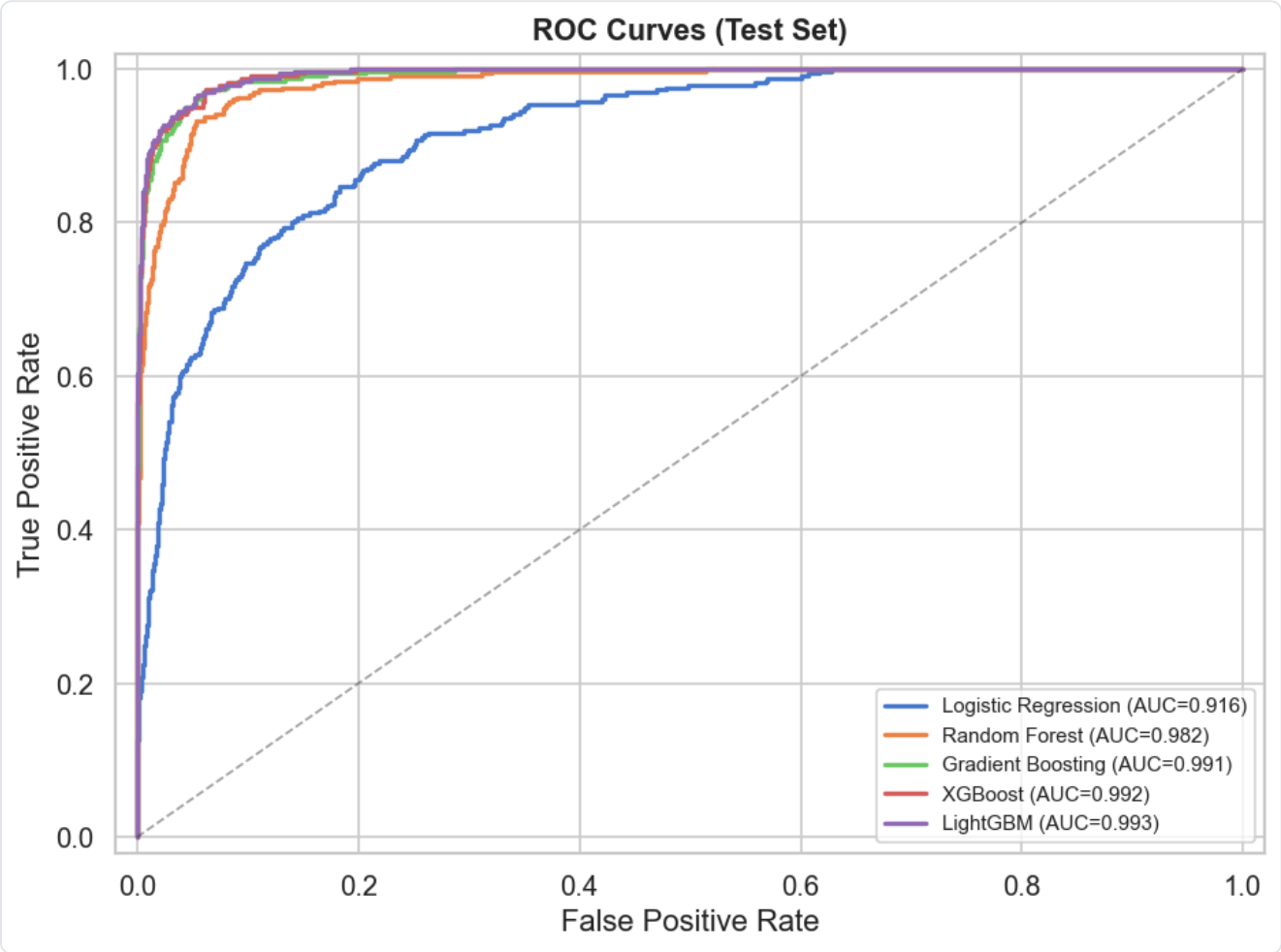
```
Best model by F1: LightGBM

              precision    recall  f1-score   support

    Existing       0.98      0.99      0.98      1701
    Attrited       0.93      0.89      0.91       325

    accuracy                           0.97      2026
```

```
   macro avg       0.96      0.94      0.95      2026
weighted avg       0.97      0.97      0.97      2026
```

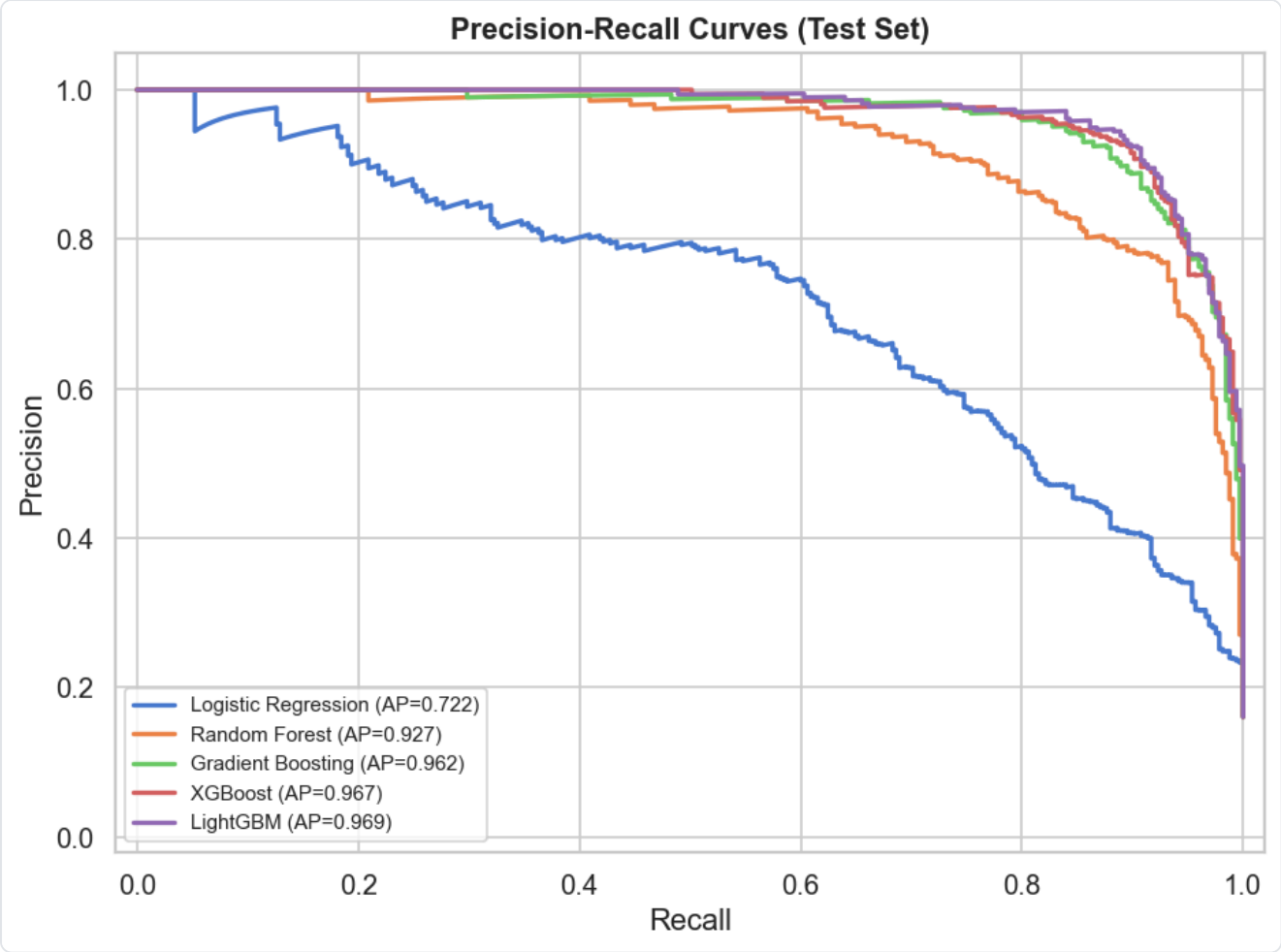
Confusion Matrices



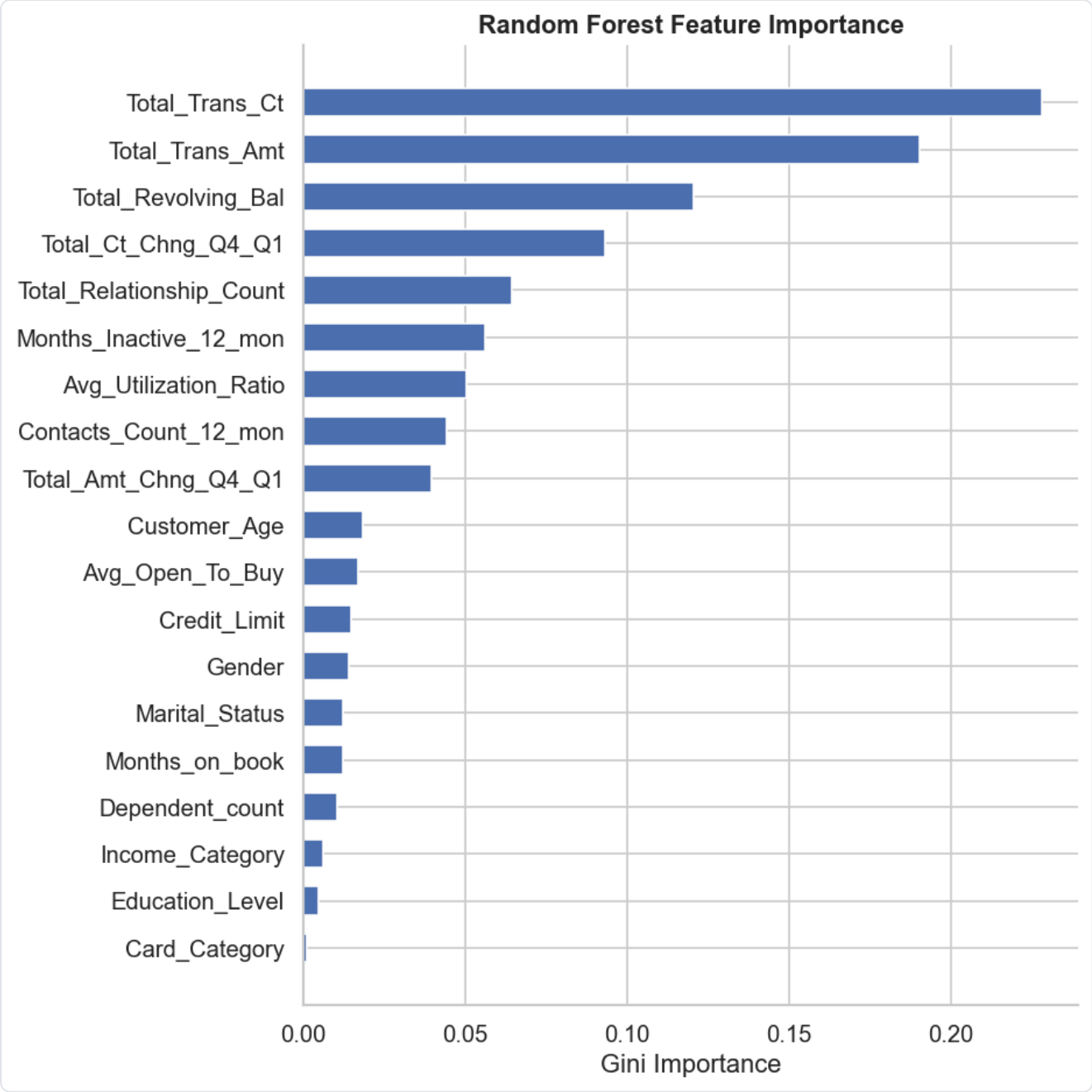
ROC Curves

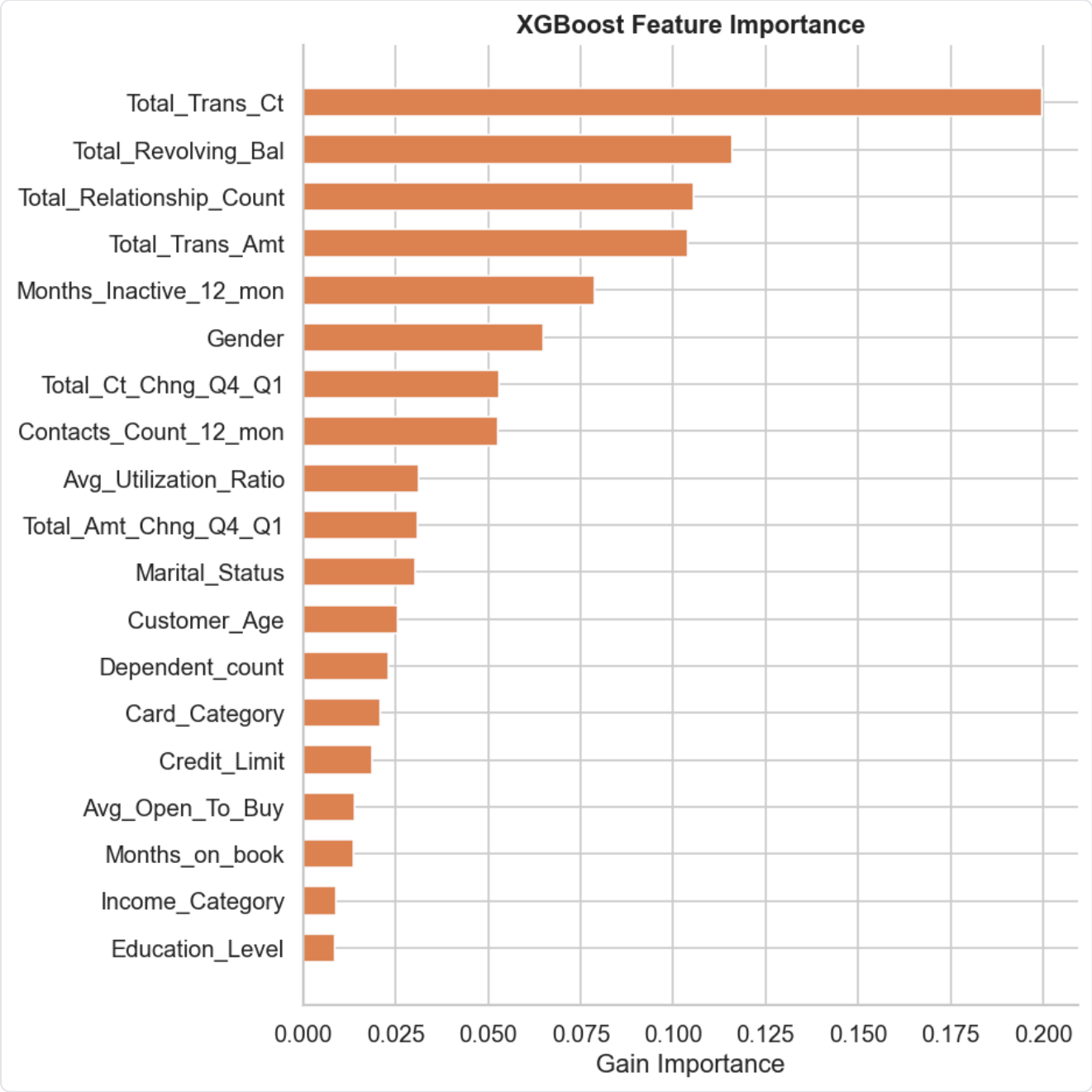


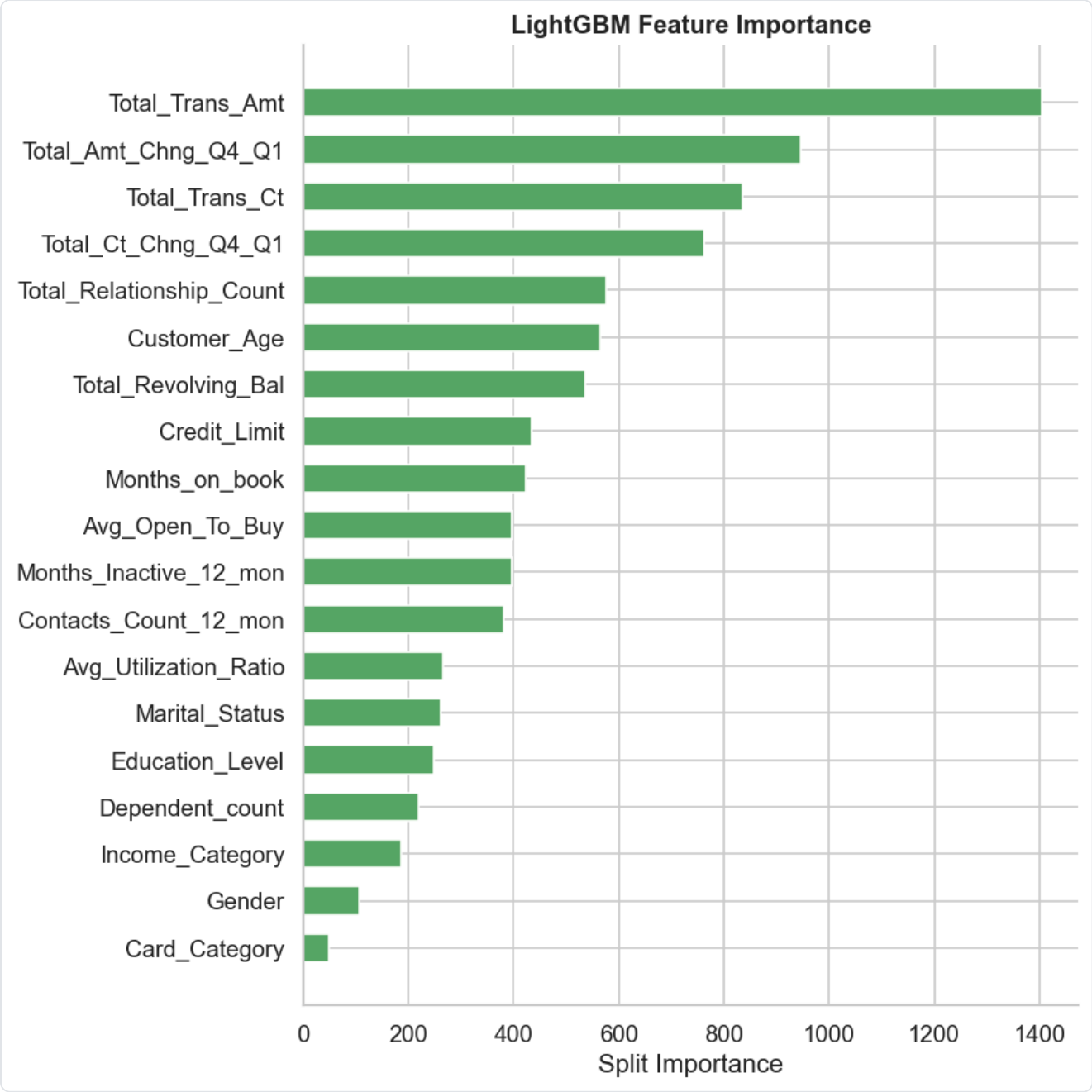
Precision-Recall Curves



Feature Importance


Random Forest Feature Importance
Total_Trans_Ct
Total_Trans_Amt
Total_Revolving_Bal
Total_Ct_Chng_Q4_Q1
Total_Relationship_Count
Months_Inactive_12_mon
Avg_Utilization_Ratio
Contacts_Count_12_mon
Total_Amt_Chng_Q4_Q1
Customer_Age
Avg_Open_To_Buy
Credit_Limit
Gender
Marital_Status
Months_on_book
Dependent_count
Income_Category
Education_Level
Card_Category
0.00
0.05
0.10
0.15
0.20
Gini Importance

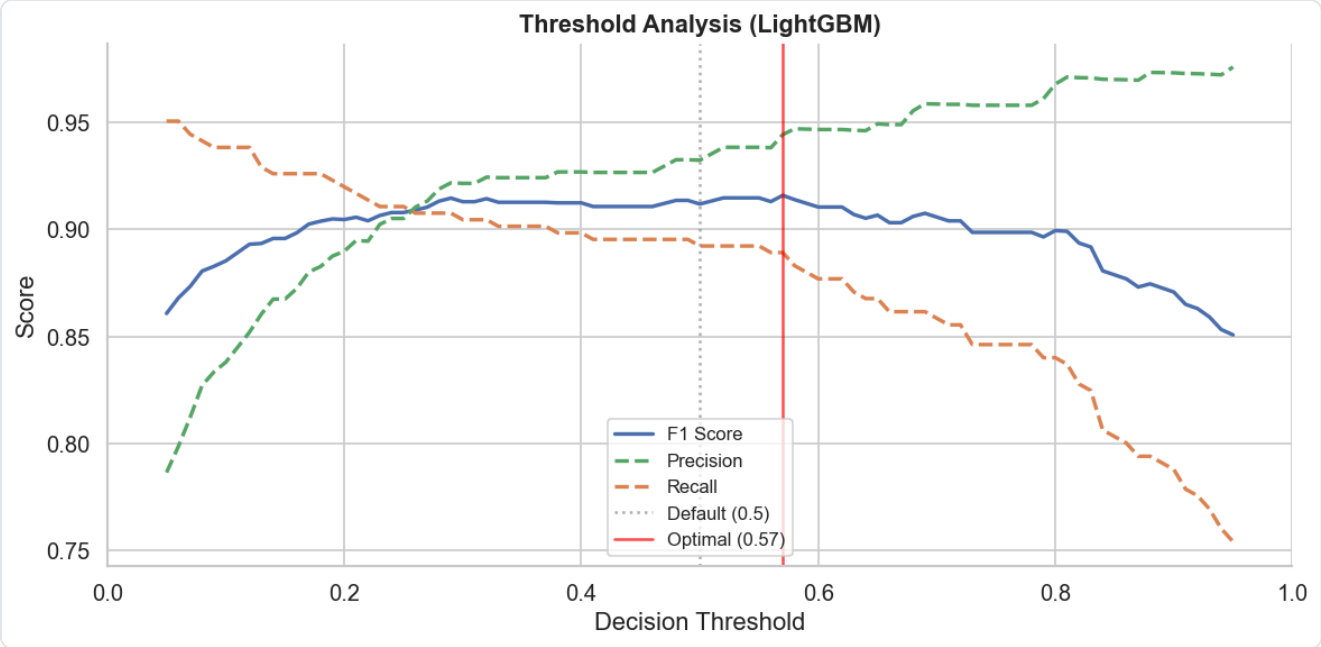

XGBoost Feature Importance
Total_Trans_Ct
Total_Revolving_Bal
Total_Relationship_Count
Total_Trans_Amt
Months_Inactive_12_mon
Gender
Total_Ct_Chng_Q4_Q1
Contacts_Count_12_mon
Avg_Utilization_Ratio
Total_Amt_Chng_Q4_Q1
Marital_Status
Customer_Age
Dependent_count
Card_Category
Credit_Limit
Avg_Open_To_Buy
Months_on_book
Income_Category
Education_Level
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200
Gain Importance



All three models agree on the top features: transaction count, transaction amount, revolving balance, and the Q4/Q1 change ratio.

6. Threshold Tuning and Retention Strategy

Threshold Analysis




```
Optimal threshold: 0.57
  F1 at optimal:   0.916
  Precision:        0.944
  Recall:           0.889
```

Lowering the threshold catches more churners at the cost of more false alarms. Where you set it depends on what a missed churner costs vs a wasted retention offer.

Cost-Benefit Simulation

Retention Cost-Benefit Analysis (Test Set)

	CUSTOMERS FLAGGED	TRUE CHURNERS CAUGHT	FALSE ALARMS	SAVED (30% RATE)	REVENUE SAVED ($)	RETENTION COST ($)	NET BENEFIT ($)
SCENARIO							
NO MODEL (DO NOTHING)	0	0	0	0	$0	$0	$0
MODEL @ 0.50 THRESHOLD	311	290	21	87	$43,500	$15,550	$27,950
MODEL @ 0.57 THRESHOLD	306	289	17	86	$43,000	$15,300	$27,700

Placeholder assumptions ($500 revenue, $50 offer, 30% save rate) to illustrate the framework. Swap in your own numbers.

Risk Tiers for Retention Targeting

Risk Tiers: Who to Call First

RISK_TIER	N_CUSTOMERS	AVG_PROB	ACTUAL_CHURN_RATE	AVG_TRANS_CT	AVG_CREDIT_LIMIT
LOW	1690	0.005000	1.5%	69.459000	8966.482000
MEDIUM	25	0.295000	36.0%	41.520000	8851.932000
HIGH	29	0.655000	58.6%	52.000000	7690.400000
CRITICAL	282	0.981000	96.8%	45.184000	8194.940000

"Critical" and "High" tier customers should be contacted first. Profile data helps reps prepare before the call.

Generated from the Credit Card Customers dataset. Full notebook and code available on GitHub.